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In fruit flies, a gene for body colour has a dominant allele for grey body, **G**, and a recessive allele for black body, **g**.

Explain how you would determine if the genotype of a grey fly is homozygous or heterozygous for body colour.

[2 marks]

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In fruit flies, males are **XY** and females are **XX**.

A cross between a grey-bodied male fly and a black-bodied female fly produced some black-bodied females. Explain how this shows that the gene for body colour is **not** sex-linked.

[1 mark]

0 4 . 3

A population of fruit flies contained 19% grey-bodied flies.

Use the Hardy–Weinberg equation to calculate the percentage of flies heterozygous for gene **G**.

Show your working.

[2 marks]

Answer _____ %

0 4 . 4

In fruit flies, a gene for wing shape has a dominant allele for curly wings, **R**, and a recessive allele for normal wings, **r**. The alleles for this gene are on a different pair of chromosomes from the gene for body colour.

Fruit flies that are homozygous dominant for curly wings do not survive beyond the embryo stage.

A curly-winged fly, homozygous for grey body colour was crossed with a curly-winged, black-bodied fly.

Complete the genetic diagram to show all the possible genotypes and the ratio of phenotypes expected to develop into adults from this cross.

[3 marks]

Phenotypes of parents Curly-winged, grey-bodied, × Curly-winged, black-bodied

Genotypes of parents _____ × _____

Genotypes of offspring _____

Phenotypes of offspring _____

Ratio of offspring _____

